Response to Reviewers' Comments

Paper ID 2147 • ICMLDE 5.0 • Procedia Computer Science

Original title: *Edge-Deployable Skin Lesion Classification: A Weighted Ensemble of MobileNetV2 and DINOv2*

Revised title: *Edge-Deployable Skin Lesion Classification with Quantized CNN and Transformer Models*

Respected Scientific Committee and Reviewers,

We sincerely thank the reviewers for their careful reading and constructive comments, which have substantially strengthened the paper. In addressing them, we carried out a detailed audit of the data split and of the model-fusion evaluation, which sharpened the focus of the contribution. The revised paper presents MobileNetV2 and DINOv2 as independent branches, analyses their probability-level fusion with a paired statistical test, and adds on-device latency, memory, and thermal measurements on a Raspberry Pi 4 for the FP32 and quantized models. All accuracies are reported transparently as validation results, together with a lesion-level overlap analysis.

Page numbers below refer to the revised manuscript.

Reviewer 1

Comment 1.1: *The abstract reports validation accuracy; independent test results should be reported.*

Response: We agree. The abstract and all results now state explicitly that they are validation results and why: a lesion-ID audit found 54.04% overlap between validation and training lesions. We also report accuracy on the 1,144 validation images whose lesions do not appear in training (MobileNetV2 86.80%, DINOv2 84.00%). A lesion-grouped split with no lesion overlap has been prepared, but retraining both models on it and running a single held-out test evaluation could not be completed within the four-day revision window. Rather than present validation figures as test performance, we scoped every claim to the validation partition, reported the non-overlapping subset as the closest available unbiased estimate, and list the held-out evaluation as the first item of future work.

Revised in the manuscript (Abstract (p. 1); Sec. 3.2 (p. 4)): "*. . . All results are validation results, because lesion-level auditing identified 54.04% overlap between validation and training images. . . .*"

Comment 1.2: *A weighted ensemble of MobileNetV2 and DINOv2 is a modest contribution.*

Response: We agree, and our audit confirmed it: the fusion gain was at most 0.40 points and was measured on the tuning data. The ensemble is no longer the contribution. The revised paper contributes (i) a controlled comparison of FP32, dynamic-range INT8, full-integer INT8, and 50% pruned MobileNetV2; (ii) on-device latency, memory, and thermal measurements on a Raspberry Pi 4; and (iii) a diagnosis of why INT8 quantization fails for DINOv2 (activation outliers in Blocks 8–11). The title now names quantization, the central subject of the revision.

Revised in the manuscript (Title; Abstract (p. 1); Sec. 3.6 (p. 4); Sec. 4.4 (p. 6)): "*. . . The two models are evaluated as independent branches; a post-hoc probability-fusion audit gave no reliable gain over MobileNetV2, so fusion is not part of the pipeline. . . .*"

Comment 1.3: *The edge-deployable claim needs latency and memory measurements on edge devices.*

Response: Done. All MobileNetV2 variants were benchmarked on a Raspberry Pi 4 Model B (4 GB, Cortex-A72, CPU only). Full INT8 TFLite ran at 13.92 ms per image (71.8 FPS) with 51.7 MiB peak memory, versus 597.59 ms and 636.9 MiB for the FP32 Keras model; an unquantized FP32 TFLite baseline (36.63 ms) separates the runtime effect from the quantization effect. Sustained 5-minute runs reached at most 78.4 °C with no throttling, and Full INT8 latency stayed constant (14.11 to 14.16 ms).

Revised in the manuscript (Sec. 4.6, Table 4 (p. 7)): "*. . . Full INT8 TFLite ran at 13.92 ms per image (71.8 FPS), 42.9× faster than FP32 Keras and 2.6× faster than unquantized FP32 TFLite, while using the least memory. . . .*"

Comment 1.4: *Ensemble weights tuned on validation data must be tested on a separate test set.*

Response: We agree. Because no separate test set was available, we removed weighted fusion from the pipeline instead of reporting a tuned result. Section 4.4 reports both audited weightings against MobileNetV2 alone and states that the equal-weight result is subject to selection bias.

Revised in the manuscript (Sec. 3.6 (p. 4)): "*. . . This fusion is an audit only and is not part of the DermaVision pipeline shown in Figure 1. . . .*"

Comment 1.5: *Comparison with published ISIC 2019 results is needed.*

Response: We considered this carefully. Published ISIC 2019 results are reported on the full eight-class task, mostly with balanced accuracy, and on splits that are rarely lesion-disjoint, whereas our study uses a three-class subset and an image-level split with documented lesion overlap. A side-by-side accuracy table would therefore compare different tasks and would overstate or understate our results. We instead state the task scope explicitly and limit our claims to a controlled comparison of compression variants under one protocol, where every number is directly comparable.

Revised in the manuscript (Abstract (p. 1); Sec. 5 (p. 8–9)): "*. . . The results therefore support comparative compression and deployment analysis, but not clinical generalization. . . .*"

Comment 1.6: *The conclusion should discuss the small 1.24-point gain.*

Response: We agree. With the corrected attribution, the best fusion gain over MobileNetV2 was 0.40 points (10 additional correct images; McNemar $p = 0.174$, not significant), and the previously reported 0.4/0.6 configuration was 0.20 points *below* MobileNetV2. The Discussion and Conclusion now state that this gain is too small and too uncertain to justify running DINOv2 alongside MobileNetV2, and that the practical result is the Full INT8 MobileNetV2.

Revised in the manuscript (Sec. 4.4 (p. 6)): "*. . . An exact two-sided McNemar test on these discordant pairs gave $p = 0.174$, so the improvement is not statistically significant at the 0.05 level; the test also assumes independent samples, which the documented lesion overlap only partly satisfies. . . .*"

Reviewer 2

Comment 2.1: *Only three classes are used; the selection should be justified.*

Response: The three classes were chosen because they cover a malignant melanocytic lesion (Melanoma), its most common benign look-alike (Nevus), and the most common non-melanocytic skin cancer (BCC). This isolates the Melanoma–Nevus confusion, which is the clinically most consequential error and the main error of our models, and keeps the class distribution comparatively balanced. The limitations of this choice are stated in the Discussion.

Revised in the manuscript (Sec. 3.1 (p. 3)): "*. . . They cover a malignant melanocytic lesion, its most common benign look-alike, and the most common non-melanocytic skin cancer, which isolates the clinically most consequential Melanoma–Nevus confusion. . . .*"

Comment 2.2: *Per-class recall, especially for melanoma, should be reported.*

Response: Done. Table 5 reports per-class precision, recall, and F1 for MobileNetV2 (Melanoma recall 87.06%), Table 3 reports Melanoma recall for every model variant, and the confusion matrix (Fig. 4) shows that the dominant error is Melanoma predicted as Nevus (78 images).

Revised in the manuscript (Sec. 4.7, Fig. 4, Table 5 (p. 8)): "*. . . BCC achieved the highest recall, while Melanoma achieved the lowest. . . .*"

Comment 2.3: *Class balance should be described.*

Response: The dataset contains 3,323 BCC, 4,522 Melanoma, and 4,600 Nevus images; the stratified split gives 2,658 / 3,618 / 3,680 training and 665 / 904 / 920 validation images. The classes are therefore close to balanced (largest-to-smallest ratio 1.38), and the stratified split preserves these proportions in both partitions.

Revised in the manuscript (Sec. 3.1 (p. 3)): "*. . . The subset contains 3,323 BCC, 4,522 Melanoma, and 4,600 Nevus images (largest-to-smallest ratio 1.38), so the classes are close to balanced. . . .*"

Comment 2.4: *Lesion-level splitting should be confirmed.*

Response: We checked this and found that the original split was image-level, not lesion-level: 54.04% of validation images belong to lesions present in training (BCC 85.11%, Melanoma 79.31%, Nevus 6.74%). We report this openly, give results on the non-overlapping subset, and have prepared a lesion-grouped split. Retraining on the lesion-grouped split is listed as future work, and this limitation is stated in the Discussion.

Revised in the manuscript (Sec. 3.2 (p. 4)): "*. . . Of the validation images, 1,345 (54.04%) belonged to lesions also represented in the training partition. . . .*"

Comment 2.5: *Statistical significance should be tested.*

Response: Done. An exact two-sided McNemar test compares MobileNetV2 with the equal-weight fusion on the same 2,489 images: fusion corrected 27 MobileNetV2 errors and introduced 17 new ones ($p = 0.174$), so the 0.40-point gain is not statistically significant at the 0.05 level. We report this explicitly, note that lesion overlap only partly satisfies the test's independence assumption, and list confidence intervals on a lesion-disjoint test set as future work.

Revised in the manuscript (Sec. 4.4 (p. 6)): "*. . . Equal-weight fusion corrected 27 MobileNetV2 errors and introduced 17 new ones, a net gain of 10 images, and raised recall in every class (BCC* +0.30*, Melanoma* +0.33*, Nevus* +0.54 *percentage points). . . .*"

Comment 2.6: *Limitations of using a three-class subset of ISIC 2019 should be stated.*

Response: Done. The Discussion states that only three of the eight ISIC 2019 categories are used, that 1,886 images lack lesion IDs, and that there is no external multi-centre validation, so the results do not support clinical generalisation.

Revised in the manuscript (Sec. 5 (p. 8–9)): "*. . . The dataset also contains only three of the eight ISIC 2019 categories, includes 1,886 images without lesion IDs, and has no external multi-centre validation. . . .*"

Reviewer 3

Comment 3.1: *DINOv2 deployment on edge devices should be discussed.*

Response: Done. DINOv2 FP32 requires 330.34 MiB and 235–373 ms per image on the development CPU. Native dynamic INT8 reduces it to 87.38 MiB and 173.54 ms but drops accuracy to 61.59%. Layer-wise diagnostics trace this to activation outliers in Blocks 8–11, a known problem for transformer quantization, and we identify quantization-aware training or distillation to a smaller ViT as the route to edge deployment.

Revised in the manuscript (Sec. 4.5 (p. 7)): "*. . . Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, a known obstacle to post-training transformer quantization, with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. . . .*"

Comment 3.2: *Grad-CAM visualisations would help.*

Response: Done. A new Grad-CAM section and figure show MobileNetV2 heatmaps for one correctly classified image of each class and for a Melanoma image predicted as Nevus. In the correct cases the activation is concentrated on the lesion; in the failure case the model also attends to the lesion but still predicts Nevus, so the error comes from interpreting the lesion rather than from attending to irrelevant regions.

Revised in the manuscript (Sec. 6, Fig. 5 (p. 9)): "*. . . In the failure case, the model also attends to the lesion but still predicts Nevus, which indicates that the error arises from how the lesion is interpreted rather than from attention to irrelevant regions. . . .*"

Comment 3.3: *Training settings should be reported.*

Response: The training settings used in this study are reported in the methodology: input size (224×224), normalization for each model, the augmentations used (rotation, zoom, translation), ImageNet and self-supervised pretraining, the classification heads, and 50 training epochs for both models, with the epoch-wise curves in Fig. 3. A full hyperparameter table could not be included within the ten-page limit.

Revised in the manuscript (Sec. 3.3 (p. 4)): "... *The verified MobileNetV2 training pipeline used rotation, zoom, and translation augmentation. ...*"

Comment 3.4: *Fairness across skin tones should be discussed.*

Response: We agree this is important. ISIC 2019 does not provide Fitzpatrick skin-type labels and contains few images of darker skin, so a reliable subgroup analysis is not possible with this dataset; colour-based skin-tone estimates from dermoscopic images are themselves unreliable. We therefore state this explicitly as a limitation and do not claim that performance generalizes across skin tones.

Revised in the manuscript (Sec. 5 (p. 8)): "... *Apart from the McNemar test of the fusion audit, no confidence intervals, skin-tone fairness evaluation, or lesion-disjoint retrained test result are available in the present study. ...*"

Comment 3.5: *Recent edge-oriented skin lesion classifiers should be cited.*

Response: The revised Related Work and methodology now cite the efficient-model and compression literature that our edge results build on: MobileNet and MobileNetV2, integer-only quantization, post-training quantization, magnitude pruning, and transformer quantization. Because our edge claim rests on direct on-device measurements rather than on comparison with other systems, which use different devices, datasets, and class sets, we positioned the work against these methods.

Revised in the manuscript (Sec. 4.5 (p. 7)): "... *Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, a known obstacle to post-training transformer quantization, with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. ...*"

Comment 3.6: *Future work could include all ISIC classes.*

Response: Done. Future work now includes extending the study to all eight ISIC 2019 categories, together with lesion-disjoint retraining, quantization-aware training, and ViT-S distillation.

Revised in the manuscript (Sec. 7 (p. 9)): "... *Extending the study to all eight ISIC 2019 categories, quantization-aware training, ViT-S distillation, out-of-distribution rejection, and imbalance-aware losses may improve the reliability of a future deployment-oriented system. ...*"

Editorial requirements

- **Figures:** all figures were produced by the authors for this manuscript from their own experiments.
- **References:** seven references were added (quantization, pruning, Grad-CAM, TensorFlow, transformer activation outliers), four previously uncited references are now cited, and the list is numbered in order of citation.

Sincerely,
Darapureddi Chetan Kumar and Jinka Venkata Aravind

Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 00 (2026) 000–000

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Machine Learning and Data Engineering

Edge-Deployable Skin Lesion Classification with Quantized CNN and Transformer Models

Darapureddi Chetan Kumar[a], Jinka Venkata Aravind[a,*]

[a]*Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham,Chennai,601103,India*

Abstract

Automated classification of skin lesions using dermoscopic images can assist in distinguishing visually similar lesion categories. This work evaluates MobileNetV2 and DINOv2 for classifying Basal Cell Carcinoma (BCC), Melanoma, and Nevus from a curated three-class subset of the ISIC 2019 dataset. The study focuses on a reproducible comparison of full-precision, quantized, and pruned models, together with a Grad-CAM explainability assessment. On the existing 2,489-image image-level validation partition, MobileNetV2 FP32 achieved 90.96% accuracy and 87.06% Melanoma recall. Full-integer INT8 TFLite conversion reduced the model size from 30.21 MiB to 2.94 MiB, while retaining 90.72% accuracy and 86.73% Melanoma recall; batch-one latency was 4.51 ms on the development CPU and 13.92 ms (71.8 FPS) on a Raspberry Pi 4 Model B, compared with 597.59 ms for the FP32 Keras model on the same device. In contrast, 50% unstructured pruning reduced accuracy to 83.49%, while native dynamic INT8 quantization of DINOv2 provided a compact 87.38 MiB representation and 173.54 ms host latency, with accuracy of 61.59% under the evaluated quantization configuration. This result identifies a clear opportunity for quantization-aware training and distillation. The two models are evaluated as independent branches; a post-hoc probability-fusion audit gave no reliable gain over MobileNetV2, so fusion is not part of the pipeline. All results are validation results, because lesion-level auditing identified 54.04% overlap between validation and training images. The results therefore support comparative compression and deployment analysis, but not clinical generalization. Grouped held-out test evaluation remains necessary before any clinical performance claim.

© 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Machine Learning and Data Engineering (ICMLDE2026).

Keywords: Skin lesion classification; Model compression; MobileNetV2; DINOv2; Edge deployment; ISIC 2019

1. Introduction

Skin lesion analysis plays an important role in the identification and assessment of dermatological conditions. BCC, Melanoma, and Nevus may exhibit visual similarities, making accurate image-based classification challenging. MobileNetV2 uses inverted residual blocks and depthwise separable convolutions to reduce computational complexity [1], whereas DINOv2 provides a transformer-based self-supervised representation [2]. Evaluating these complementary architectures is useful for understanding the accuracy–efficiency trade-off in resource-constrained applications.

This study uses a curated three-class subset of ISIC 2019. The selected classes provide a focused evaluation of BCC, Melanoma, and Nevus, with particular attention to the clinically important confusion between Melanoma and Nevus. The experiments compare FP32 models, MobileNetV2 Dynamic-Range and Full INT8 TFLite models, a 50%

* Corresponding author.
E-mail address: jv_aravind@ch.amrita.edu
1877-0509 © 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Machine Learning and Data Engineering (ICMLDE2026).

unstructured-pruned MobileNetV2, and DINOv2 dynamic INT8 diagnostics. The two architectures are evaluated as independent branches; a post-hoc probability-fusion audit is reported only to explain why fusion is excluded from the pipeline.

The main contributions of this work are summarized as follows:

- Development of a three-class skin lesion classification framework for BCC, Melanoma, and Nevus.
- Evaluation of MobileNetV2 and DINOv2 using 50-epoch training experiments.
- Analysis of training and validation accuracy and loss using the recorded epoch-wise experimental histories.
- Independent evaluation of the two branches, with a post-hoc audit showing that weighted probability fusion does not provide a reliable improvement over MobileNetV2.
- Quantitative comparison of model accuracy, class-level recall, storage size, and latency on a development CPU and a Raspberry Pi 4 across the evaluated model variants.
- Analysis of the trade-off between Full INT8 quantization and 50% unstructured pruning for MobileNetV2.
- Diagnosis of DINOv2 activation outliers that define the optimization target for quantization-aware training and distillation.
- Explicit auditing of lesion overlap and transparent separation of comparative validation evidence from independent clinical test evidence.
- Integration of the MobileNetV2 inference pipeline into a Streamlit-based application, and on-device benchmarking of the FP32 and quantized MobileNetV2 variants on a Raspberry Pi 4, including latency, memory, and sustained thermal behaviour.

2. Related Work

2.1. Convolutional Approaches to Skin Lesion Classification

Deep convolutional neural networks have demonstrated strong performance in automated skin lesion analysis. Esteva et al. demonstrated dermatologist-level skin cancer classification using deep neural networks, highlighting the potential of deep learning for dermatological image analysis [3].

Convolutional neural networks such as ResNet and MobileNet have subsequently been applied to image classification problems because of their ability to learn hierarchical visual features [4, 5]. MobileNetV2 improves computational efficiency using inverted residual structures and linear bottlenecks [1], making it suitable for applications requiring lightweight models.

2.2. Transformer-Based Visual Representation Learning

Vision Transformers provide an alternative approach to image analysis by modelling relationships between image regions using attention mechanisms [6]. Self-supervised learning approaches further improve the ability of models to learn transferable visual representations without requiring large quantities of labelled data.DINO introduced self-distillation for learning visual representations [7], while DINOv2 extended this approach to produce robust and transferable visual features suitable for multiple downstream computer vision tasks [2]. Such representations can be adapted for medical image classification through fine-tuning and task-specific classification layers.

2.3. Ensemble Learning

Ensemble learning combines predictions generated by multiple classifiers to improve overall performance. The potential benefit of an ensemble depends on the diversity of the individual models and the differences in their prediction errors [8, 9]. Combining convolutional and transformer-based architectures may provide complementary information because the two models learn visual representations using different architectural mechanisms. This work audits whether weighted probability-level fusion of MobileNetV2 and DINOv2 offers a reliable gain over the stronger single model.

3. Proposed Methodology

Figure 1 presents the overall architecture of the proposed skin lesion classification framework. The 12,445 ISIC 2019 images (BCC, Melanoma, and Nevus) are resized to 224×224, augmented, and normalized to the range 0–1, and are then divided by a stratified image-level split into 9,956 training and 2,489 validation images. The preprocessed image is provided independently to two branches. The MobileNetV2 branch applies an ImageNet-pretrained [10] backbone, global average pooling, and dense layers of 256, 64, and 3 units. The DINOv2 branch converts the image into 256 patch embeddings and a CLS token, encodes them with 12 transformer blocks, and classifies the CLS representation with a linear head (768 → 3). For each branch, the figure reports the baseline FP32 evaluation results and the subsequent optimization experiments. Full INT8 TFLite quantization [11] of MobileNetV2 retained 90.72% accuracy and is identified as the recommended edge candidate, whereas INT8 quantization of DINOv2 reduced accuracy to 61.59%. Grad-CAM [12] is applied to MobileNetV2 to visualize the lesion regions that drive its predictions.

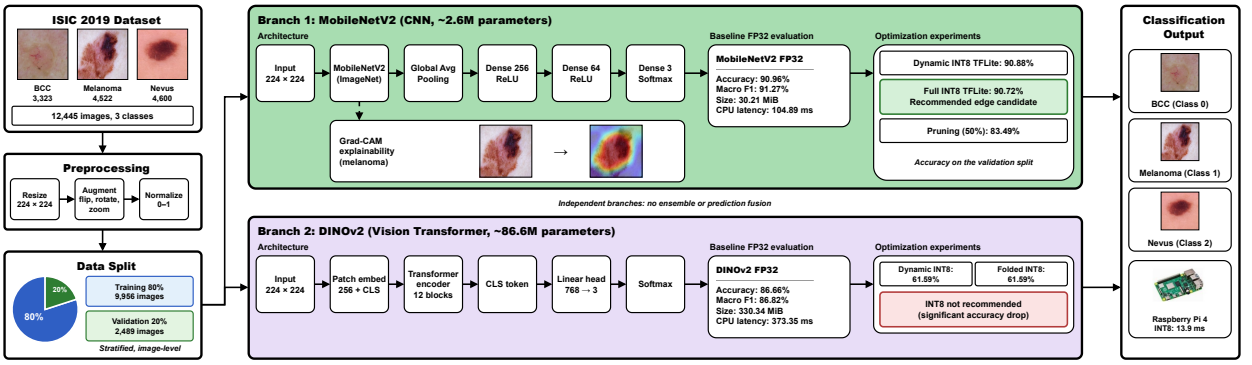


Fig. 1. Overall architecture of the DermaVision framework: data preparation, the MobileNetV2 and DINOv2 branches with their baseline FP32 evaluation and optimization experiments, Grad-CAM explainability, and the three output classes. Accuracy values refer to the 2,489-image validation split.

3.1. Dataset

The experimental study was conducted using a selected three-class subset of the ISIC 2019 skin lesion dataset [13], which combines images from the HAM10000 [14], BCN20000 [15], and ISIC 2017 challenge [16] collections. The three categories considered in this work were Basal Cell Carcinoma (BCC), Melanoma, and Nevus. They cover a malignant melanocytic lesion, its most common benign look-alike, and the most common non-melanocytic skin cancer, which isolates the clinically most consequential Melanoma–Nevus confusion. The subset contains 3,323 BCC, 4,522 Melanoma, and 4,600 Nevus images (largest-to-smallest ratio 1.38), so the classes are close to balanced. These categories include malignant and benign skin lesion types and establish a multi-class classification task. The class labels were maintained consistently throughout the experiments to ensure correct alignment between the ground-truth labels and the prediction probabilities generated by MobileNetV2 and DINOv2.

The class mapping used in this work is shown in Table 1.

Table 1. Class mapping used for skin lesion classification.

Class Label	Skin Lesion Category
0	BCC
1	Melanoma
2	Nevus

For final comparative evaluation, an image-level validation partition containing 2,489 images was used: 665 BCC, 904 Melanoma, and 920 Nevus images. The same class-label mapping and image ordering were maintained while generating predictions from MobileNetV2 and DINOv2 so that the predictions of both models could be compared image by image. This partition is not treated as an independent test set.

3.2. Data Splitting and Leakage Audit

The legacy evaluation protocol used a stratified image-level 80:20 split, producing 9,956 training images and 2,489 validation images. A subsequent lesion-ID audit identified 5,502 unique known lesions and 1,886 images without a lesion ID. Of the validation images, 1,345 (54.04%) belonged to lesions also represented in the training partition. The overlap was particularly high for BCC (85.11% of validation images) and Melanoma (79.31%), while the corresponding Nevus overlap was 6.74%.

The overlap materially changes the interpretation of the results. On the clean non-overlapping subset of 1,144 validation images, MobileNetV2 accuracy decreased from 90.96% to 86.80%, DINOv2 accuracy decreased from 86.66% to 84.00%, and the post-hoc equal-weight fusion decreased from 91.36% to 87.15%. Lesion-grouped Revision V1 split manifests have been prepared, but valid retrained models and held-out test predictions are not yet available. Accordingly, the reported results quantify relative model and compression behaviour on the legacy validation partition.

3.3. Data Preprocessing

The input dermoscopic images were resized to 224×224 pixels. MobileNetV2 inputs were converted to RGB and scaled by $1/255$ to the range $[0, 1]$. DINOv2 inputs were normalized using the ImageNet mean and standard deviation. The verified MobileNetV2 training pipeline used rotation, zoom, and translation augmentation. Horizontal flipping and brightness augmentation are not claimed for the reported run. After preprocessing, the images were converted into the required tensor representation and supplied to the respective models.

3.4. MobileNetV2-Based Classification

MobileNetV2 was employed as the convolutional neural network for the skin lesion classification task. The architecture is designed to provide computational efficiency through inverted residual blocks and depthwise separable convolutions [1]. The pretrained MobileNetV2 architecture was adapted for classification of the three target categories: BCC, Melanoma, and Nevus. The model was trained for 50 epochs. During training, training accuracy, validation accuracy, training loss, and validation loss were recorded for every epoch. The recorded 50-epoch history was subsequently used to generate the corresponding performance curves. The recorded epoch history reached 90.12% validation accuracy at epoch 50; the saved FP32 checkpoint achieved 90.96% on the consolidated 2,489-image evaluation partition.

3.5. DINOv2-Based Classification

DINOv2 was employed as the transformer-based model for the skin lesion classification task. The pretrained model was adapted to classify the three target categories: BCC, Melanoma, and Nevus. The DINOv2 classifier was trained for 50 epochs. Training accuracy, validation accuracy, training loss, and validation loss were recorded throughout the training process. The recorded epoch history reached 86.82% validation accuracy at epoch 35; the saved FP32 checkpoint achieved 86.66% on the consolidated evaluation partition.

3.6. Post-hoc Fusion Audit

To test whether the two branches carry complementary information, their probability vectors were combined after training, as defined in Equation (1):

$$P_{\text{fused}} = w_M P_M + w_D P_D, \qquad w_M + w_D = 1, \qquad \hat{y} = \arg\max(P_{\text{fused}}) \tag{1}$$

Here, P_M and P_D are the probability vectors of MobileNetV2 and DINOv2, and w_M and w_D are their weights. This fusion is an audit only and is not part of the DermaVision pipeline shown in Figure 1. Its results are reported in Section 4.4.

3.7. System Implementation and Edge Deployment

The developed skin lesion classification framework was integrated into a Streamlit-based application. The application accepts a dermoscopic skin lesion image as input and performs preprocessing before the image is provided to the trained classification pipeline. The application workflow consists of three main stages: image quality assessment, model prediction, and presentation of the classification result. After an image is uploaded, the system first performs an image quality assessment. The accepted image is then processed by the MobileNetV2 TFLite model, which generates the final prediction. The output interface displays the predicted skin lesion category together with the associated confidence scores. This enables the user to observe the relative confidence assigned to the three classes: BCC, Melanoma, and Nevus. Figure 2 illustrates the implemented application workflow, including the diagnostic interface, image quality assessment stage, and final prediction output.

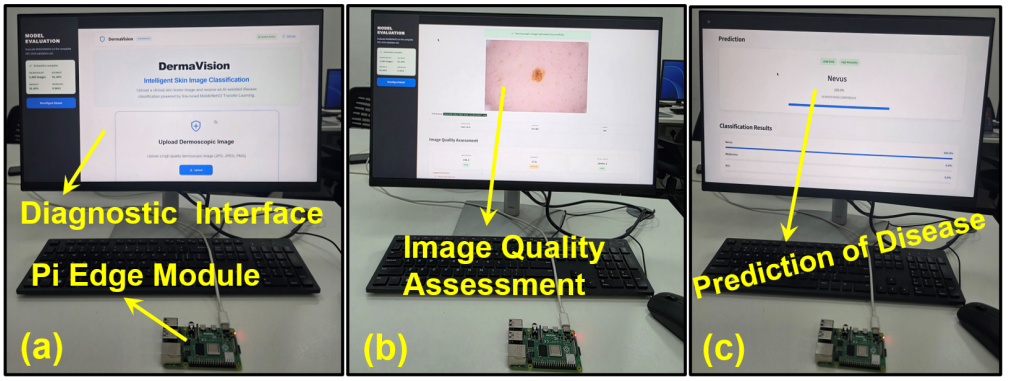


Fig. 2. Application workflow showing (a) the diagnostic interface and Pi Edge Module, (b) the image quality assessment stage, and (c) the final disease prediction with confidence scores.

The application currently uses the Dynamic-Range TFLite MobileNetV2 model. A separate Full INT8 TFLite inference module is available and is the recommended edge candidate; its Raspberry Pi 4 latency, memory, and thermal behaviour are reported in Section 4.6.

4. Experimental Results and Analysis

4.1. Experimental Setup

Both MobileNetV2 and DINOv2 were trained for 50 epochs on the selected three-class skin lesion dataset. During training, training accuracy, validation accuracy, training loss, and validation loss were recorded for every epoch. The complete recorded histories were used to generate the performance curves. The consolidated evaluation compared seven variants using accuracy, precision, recall, F1-score, model size, and batch-one host-CPU latency. On-device measurements on a Raspberry Pi 4 are reported separately in Section 4.6.

4.2. Individual Model Performance

MobileNetV2 was trained for 50 epochs, and its training and validation performance was monitored throughout the training process. The recorded curves show the progression of both accuracy and loss across the epochs. The epoch history reached 90.12% at epoch 50, while the saved FP32 checkpoint achieved 90.96% on the consolidated validation partition. Figure 3(a,b) presents the recorded curves for MobileNetV2.

DINOv2 was also trained for 50 epochs, and its training and validation performance was observed across the training process. The epoch history reached its highest value of 86.82% at epoch 35, while the saved FP32 checkpoint achieved 86.66% on the consolidated validation partition. Figure 3(c,d) shows the recorded curves for DINOv2.

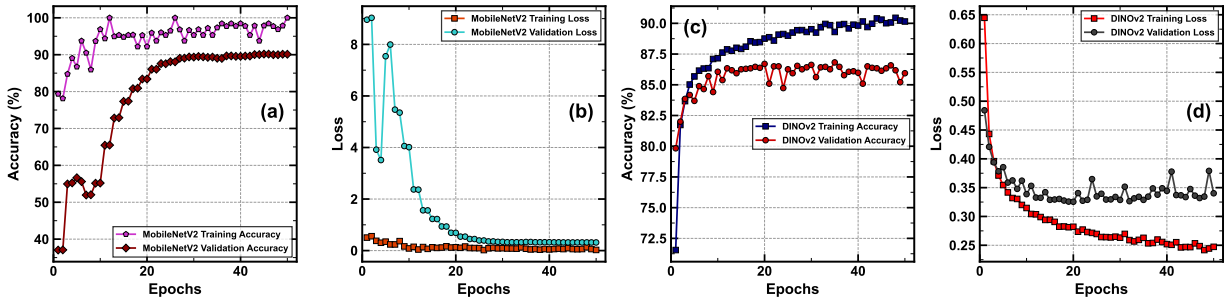


Fig. 3. Recorded training and validation curves over 50 epochs: (a) MobileNetV2 accuracy, (b) MobileNetV2 loss, (c) DINOv2 accuracy, and (d) DINOv2 loss. MobileNetV2 reached its highest validation accuracy of 90.16% at epoch 45 (90.12% at epoch 50); DINOv2 reached 86.82% at epoch 35.

4.3. Comparison of Individual Models

The consolidated results show that MobileNetV2 FP32 achieved 90.96% accuracy, 91.27% Macro F1, and 87.06% Melanoma recall. DINOv2 FP32 achieved 86.66% accuracy, 86.82% Macro F1, and 79.09% Melanoma recall. These values are validation results from the image-level partition described in Section 3.2.

Because MobileNetV2 achieved the higher accuracy at a fraction of the size and latency of DINOv2, it is treated as the primary deployment branch, while DINOv2 is retained as the transformer comparison branch.

4.4. Post-hoc Fusion Audit Results

Table 2 compares MobileNetV2 with the two audited fusion configurations. The previously reported configuration ($w_M = 0.4$, $w_D = 0.6$) achieved 90.76% accuracy and 85.18% Melanoma recall, both below MobileNetV2 alone. The equal-weight configuration ($w_M = 0.5$, $w_D = 0.5$) exceeded MobileNetV2 by only 0.40 percentage points, with weights evaluated on the same validation predictions and on a partition with documented lesion overlap. Because this gain is small, unverified on held-out data, and requires running the 330.34 MiB DINOv2 model alongside MobileNetV2, fusion is not adopted, and the remaining analysis focuses on the independent branches.

The two branches make partly different errors: DINOv2 correctly classified 110 images that MobileNetV2 misclassified, but misclassified 217 images that MobileNetV2 classified correctly. Equal-weight fusion corrected 27 MobileNetV2 errors and introduced 17 new ones, a net gain of 10 images, and raised recall in every class (BCC +0.30, Melanoma +0.33, Nevus +0.54 percentage points). An exact two-sided McNemar test on these discordant pairs gave $p = 0.174$, so the improvement is not statistically significant at the 0.05 level; the test also assumes independent samples, which the documented lesion overlap only partly satisfies. On the 1,144 non-overlapping validation images the gain was 0.35 percentage points (87.15% versus 86.80%).

Table 2. Post-hoc fusion audit compared with MobileNetV2 alone on the validation partition.

Model / Configuration	**Accuracy**	**Macro F1**	**Melanoma Recall**
MobileNetV2 FP32 (adopted)	90.96%	91.27%	87.06%
Fusion audit ($w_M = 0.5$, $w_D = 0.5$)	91.36%	91.67%	87.39%
Fusion audit ($w_M = 0.4$, $w_D = 0.6$)	90.76%	91.01%	85.18%

4.5. Compression and Development-CPU Deployment Analysis

Table 3 summarizes the principal compression and deployment measurements. Full INT8 TFLite MobileNetV2 reduced storage by 90.27% relative to the FP32 checkpoint and retained accuracy within 0.24 percentage points. It also achieved 4.51 ms batch-one latency on the benchmark host. In contrast, 50% unstructured magnitude pruning [17] reduced accuracy by 7.47 percentage points and Melanoma recall by 13.61 percentage points without providing a dense CPU speedup. DINOv2 FP32 required 330.34 MiB on disk and showed 235.09–373.35 ms host latency. Native

dynamic INT8 reduced DINOv2 storage to 87.38 MiB and latency to 173.54 ms, establishing a compact optimization baseline of 61.59% accuracy and 49.67% Melanoma recall for future quantization-aware improvement.

Table 3. Consolidated validation and development-CPU deployment measurements.

Model	Accuracy	Macro F1	Mel. Recall	Size / Latency
MobileNetV2 FP32	90.96%	91.27%	87.06%	30.21 MiB / 104.89 ms
MobileNetV2 Dynamic TFLite	90.88%	91.20%	85.62%	2.74 MiB / 27.76 ms
MobileNetV2 Full INT8 TFLite	**90.72%**	**91.03%**	**86.73%**	**2.94 MiB / 4.51 ms**
MobileNetV2 50% Pruned	83.49%	83.55%	73.45%	30.21 MiB / 101.54 ms
DINOv2 FP32	86.66%	86.82%	79.09%	330.34 MiB / 235–373 ms
DINOv2 Dynamic INT8	61.59%	57.54%	49.67%	87.38 MiB / 173.54 ms

The DINOv2 INT8 result provides a useful optimization diagnosis. Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, a known obstacle to post-training transformer quantization [18, 19], with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. These observations identify the principal target for quantization-aware training, per-channel scaling, calibration refinement, or ViT-S distillation before deployment.

4.6. Raspberry Pi 4 On-Device Benchmark

All MobileNetV2 variants were benchmarked on a Raspberry Pi 4 Model B (Broadcom BCM2711, four Cortex-A72 cores at up to 1.8 GHz, 4 GB RAM, 64-bit Raspberry Pi OS, Python 3.13.5) without any hardware accelerator. TFLite models were executed with ai-edge-litert 2.3.0 (XNNPACK, four threads) and the FP32 Keras model with TensorFlow 2.21.0 [20], using the same batch-one protocol as on the development CPU (50 warm-up and 200 timed runs on a preloaded tensor). The model files were byte-identical to those evaluated on the development CPU, so the accuracy values in Table 3 apply unchanged; every variant predicted the same class for the benchmark image. An unquantized FP32 TFLite conversion was also measured to separate the effect of the runtime from that of quantization. Table 4 summarizes the results.

Table 4. Raspberry Pi 4 Model B results for MobileNetV2 (batch one, four threads). Peak RSS is the memory of the whole Python process; Max temp. is the peak SoC temperature during a 300 s continuous run.

Variant	Disk	Latency (ms)	FPS	Peak RSS	Max temp.
FP32 Keras	30.21 MiB	597.59 ± 9.86	1.67	636.9 MiB	55.0 °C
FP32 TFLite (unquantized)	9.79 MiB	36.63 ± 0.34	27.30	69.5 MiB	74.5 °C
Dynamic-range INT8 TFLite	2.74 MiB	41.72 ± 3.42	23.97	59.2 MiB	72.0 °C
Full INT8 TFLite	**2.94 MiB**	**13.92 ± 0.20**	**71.83**	**51.7 MiB**	**78.4 °C**

Full INT8 TFLite ran at 13.92 ms per image (71.8 FPS), 42.9× faster than FP32 Keras and 2.6× faster than unquantized FP32 TFLite, while using the least memory. Most of the FP32 Keras latency is runtime overhead: converting the same FP32 weights to TFLite reduced latency from 597.59 ms to 36.63 ms. Dynamic-range quantization reduced storage to 2.74 MiB but was slightly slower than FP32 TFLite (41.72 ms), because its weights are dequantized during inference [21]; only full-integer quantization delivered a speedup on the Cortex-A72. With a single thread, Full INT8 required 45.86 ms and Dynamic-range 65.11 ms. Including decoding and preprocessing of a 1024×1024 PNG input, the Full INT8 end-to-end latency was 106.08 ms, so image decoding rather than inference dominates the on-device pipeline.

Sustained behaviour was measured by running each variant continuously for 300 s from a starting temperature below 50 °C, while sampling the SoC temperature, ARM clock, and firmware throttling flags every 2 s. No throttling or under-voltage flag was raised in any run. Full INT8, which kept all four cores busy and completed 21,118 inferences, reached the highest temperature (78.4 °C), yet its mean latency was unchanged between the first and last 10% of the run (14.11 and 14.16 ms). FP32 Keras reached only 55.0 °C because it did not saturate the CPU.

4.7. Confusion Matrix and Error Analysis

Figure 4 presents the confusion matrix of MobileNetV2 FP32 for the three target classes: BCC, Melanoma, and Nevus. The diagonal elements represent correctly classified samples, while the off-diagonal elements represent misclassifications.

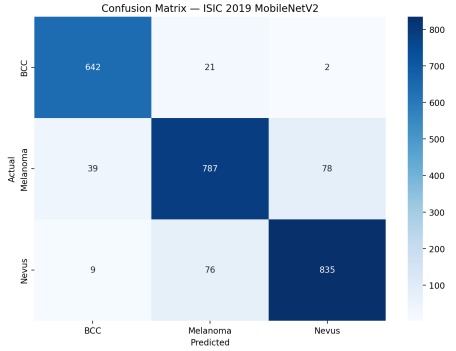


Fig. 4. Confusion matrix of MobileNetV2 FP32 for BCC, Melanoma, and Nevus classification on the 2,489-image validation partition.

For the BCC class, 642 images were correctly classified, 21 were predicted as Melanoma, and 2 were predicted as Nevus. For the Melanoma class, 787 images were correctly classified, 39 were predicted as BCC, and 78 were predicted as Nevus. For the Nevus class, 835 images were correctly classified, 9 were predicted as BCC, and 76 were predicted as Melanoma.

Table 5 presents the class-level precision, recall, and F1-score values derived from the confusion matrix.

Table 5. Class-level performance of MobileNetV2 FP32.

Class	Support	Precision	Recall	F1-score
BCC	665	93.04%	96.54%	94.76%
Melanoma	904	89.03%	87.06%	88.03%
Nevus	920	91.26%	90.76%	91.01%
Macro avg	2,489	91.11%	91.45%	91.27%

BCC achieved the highest recall, while Melanoma achieved the lowest. The confusion between Melanoma and Nevus is the main source of classification error: 78 Melanoma images were predicted as Nevus, and 76 Nevus images were predicted as Melanoma.

5. Discussion

The experimental results demonstrate that MobileNetV2 provides the strongest accuracy–efficiency trade-off among the evaluated variants. Its FP32 checkpoint achieved 90.96% validation accuracy and 87.06% Melanoma recall, while Full INT8 conversion retained 90.72% accuracy and 86.73% Melanoma recall with a substantially smaller model and lower host-CPU latency. On a Raspberry Pi 4 it ran at 71.8 FPS, which confirms Full INT8 MobileNetV2 as a practical edge model.

The post-hoc fusion audit tested whether combining the two branches improves accuracy. The 0.4/0.6 configuration (90.76%) did not exceed MobileNetV2, and the equal-weight configuration exceeded it by only 0.40 percentage points, with weights evaluated on the same validation predictions. The lesion-overlap audit further showed a 7.79 percentage-point gap between overlapping and clean validation images for the equal-weight fusion. Given this small, unverified gain and the added cost of running DINOv2, the pipeline keeps the two branches independent.

The class-level analysis shows that Melanoma remains the most important error category, particularly because Melanoma images are confused with Nevus. The 50% pruning experiment further shows that reducing parameter magnitudes without a suitable sparse execution strategy can harm this clinically relevant class. The DINOv2 INT8

experiment complements this result by identifying activation calibration as the main optimization bottleneck for transformer compression.

The framework is integrated into a Streamlit application, but the current application uses Dynamic-Range TFLite rather than the Full INT8 candidate. Apart from the McNemar test of the fusion audit, no confidence intervals, skin-tone fairness evaluation, or lesion-disjoint retrained test result are available in the present study. The dataset also contains only three of the eight ISIC 2019 categories, includes 1,886 images without lesion IDs, and has no external multi-centre validation. These limitations prevent clinical generalization.

6. Grad-CAM Explainability

Figure 5 shows Grad-CAM [12] heatmaps computed on the final convolutional layer of MobileNetV2 (7×7×1280) for one correctly classified image of each class and for a Melanoma image predicted as Nevus. In the correct cases, the activation is concentrated on the lesion rather than on the surrounding skin, hair, or imaging artefacts. In the failure case, the model also attends to the lesion but still predicts Nevus, which indicates that the error arises from how the lesion is interpreted rather than from attention to irrelevant regions. Grad-CAM is a coarse post-hoc attribution and does not establish clinical explainability.

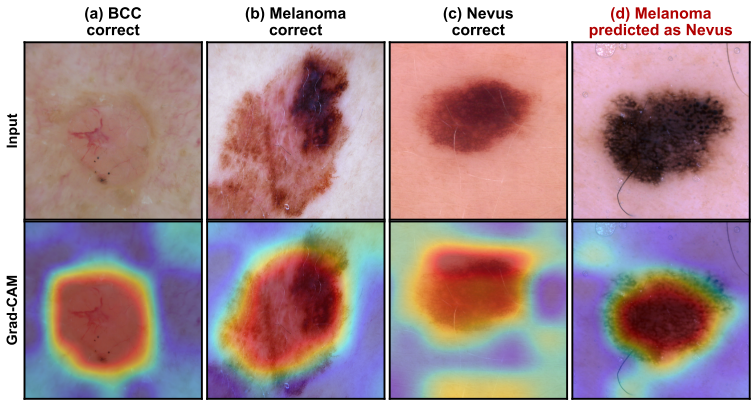


Fig. 5. Grad-CAM heatmaps of MobileNetV2 FP32: (a)–(c) correctly classified BCC, Melanoma, and Nevus images; (d) a Melanoma image predicted as Nevus. Top: input; bottom: heatmap overlay.

7. Conclusion and Future Work

This work presented a comparative study of MobileNetV2 and DINOv2 for three-class skin lesion classification. On the legacy image-level validation partition, MobileNetV2 FP32 achieved 90.96% accuracy, while Full INT8 TFLite reduced storage to 2.94 MiB and retained 90.72% accuracy with 86.73% Melanoma recall. A post-hoc fusion audit showed no reliable gain over MobileNetV2 alone, so the two branches are kept independent. All results were obtained on a validation partition with 54.04% lesion overlap and therefore cannot be interpreted as independent test performance. The experiments established Full INT8 MobileNetV2 as the current practical edge baseline and identified the activation-calibration strategy required to improve DINOv2 compression.

The main practical conclusion is that Full INT8 MobileNetV2 is the most promising candidate for immediate edge evaluation, while the DINOv2 INT8 result provides a measurable starting point for quantization-aware redesign and distillation. On a Raspberry Pi 4 it reached 13.92 ms per image (71.8 FPS) with 51.7 MiB peak memory, compared with 597.59 ms for FP32 Keras. The present results do not establish clinical generalization.

Future work will retrain both branches using the prepared zero-overlap lesion-grouped split, freeze any Melanoma safety threshold on validation data, re-test fusion on the lesion-disjoint split, and perform one final held-out test evaluation. Further work should include paired statistical tests, confidence intervals, power measurements and testing on additional edge devices, attention rollout for DINOv2, skin-tone fairness analysis, calibration, and comparison with external datasets. Extending the study to all eight ISIC 2019 categories, quantization-aware training, ViT-S distillation,

out-of-distribution rejection, and imbalance-aware losses may improve the reliability of a future deployment-oriented system.

Acknowledgment

The authors would like to thank Amrita Vishwa Vidyapeetham, Chennai Campus, for providing the computing infrastructure and academic support for this work. The authors also acknowledge the AI DISC Lab and the NEGCES Lab for their support and facilities during the development and evaluation of this work.

References

[1] Sandler, Mark, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. (2018) "MobileNetV2: Inverted residuals and linear bottlenecks." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 4510–4520.

[2] Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, et al. (2024) "DINOv2: Learning robust visual features without supervision." *Transactions on Machine Learning Research*.

[3] Esteva, Andre, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. (2017) "Dermatologist-level classification of skin cancer with deep neural networks." *Nature* **542** (7639): 115–118.

[4] He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. (2016) "Deep residual learning for image recognition." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 770–778.

[5] Howard, Andrew G., Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. (2017) "MobileNets: Efficient convolutional neural networks for mobile vision applications." *arXiv preprint* arXiv:1704.04861.

[6] Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, et al. (2021) "An image is worth 16x16 words: Transformers for image recognition at scale." *International Conference on Learning Representations*.

[7] Caron, Mathilde, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. (2021) "Emerging properties in self-supervised vision transformers." *Proceedings of the IEEE/CVF International Conference on Computer Vision*: 9650–9660.

[8] Kittler, Josef, Mohamad Hatef, Robert P. W. Duin, and Jiri Matas. (1998) "On combining classifiers." *IEEE Transactions on Pattern Analysis and Machine Intelligence* **20** (3): 226–239.

[9] Dietterich, Thomas G. (2000) "Ensemble methods in machine learning." *Multiple Classifier Systems, Lecture Notes in Computer Science* **1857**: 1–15.

[10] Deng, Jia, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. (2009) "ImageNet: A large-scale hierarchical image database." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 248–255.

[11] Jacob, Benoit, Skirmantas Kligys, Bo Chen, Menglong Zhu, Matthew Tang, Andrew Howard, Hartwig Adam, and Dmitry Kalenichenko. (2018) "Quantization and training of neural networks for efficient integer-arithmetic-only inference." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 2704–2713.

[12] Selvaraju, Ramprasaath R., Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. (2017) "Grad-CAM: Visual explanations from deep networks via gradient-based localization." *Proceedings of the IEEE International Conference on Computer Vision*: 618–626.

[13] Larxel, "Skin Lesion Images for Melanoma Classification," Kaggle Dataset, 2020. Available: https://www.kaggle.com/datasets/andrewmvd/isic-2019.

[14] Tschandl, Philipp, Cliff Rosendahl, and Harald Kittler. (2018) "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions." *Scientific Data* **5**: 180161.

[15] Combalia, Marc, Noel C. F. Codella, Veronica Rotemberg, Brian Helba, Veronica Vilaplana, Ofer Reiter, Cristina Carrera, et al. (2019) "BCN20000: Dermoscopic lesions in the wild." *arXiv preprint* arXiv:1908.02288.

[16] Codella, Noel C. F., David Gutman, M. Emre Celebi, Brian Helba, Michael A. Marchetti, Stephen W. Dusza, Aadi Kalloo, et al. (2018) "Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC)." *IEEE 15th International Symposium on Biomedical Imaging*: 168–172.

[17] Han, Song, Jeff Pool, John Tran, and William J. Dally. (2015) "Learning both weights and connections for efficient neural networks." *Advances in Neural Information Processing Systems* **28**: 1135–1143.

[18] Bondarenko, Yelysei, Markus Nagel, and Tijmen Blankevoort. (2021) "Understanding and overcoming the challenges of efficient transformer quantization." *Proceedings of the Conference on Empirical Methods in Natural Language Processing*: 7947–7969.

[19] Dettmers, Tim, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. (2022) "LLM.int8(): 8-bit matrix multiplication for transformers at scale." *Advances in Neural Information Processing Systems* **35**: 30318–30332.

[20] Abadi, Martín, Paul Barham, Jianmin Chen, Zhifeng Chen, Andy Davis, Jeffrey Dean, Matthieu Devin, et al. (2016) "TensorFlow: A system for large-scale machine learning." *12th USENIX Symposium on Operating Systems Design and Implementation*: 265–283.

[21] Krishnamoorthi, Raghuraman. (2018) "Quantizing deep convolutional networks for efficient inference: A whitepaper." *arXiv preprint* arXiv:1806.08342.